Dark Matter under the LVC Lens v1: A Coordinate Rewriting of Ω_m, a Universal Conservation Identity, and the Soliton-Charge Structure of the Particle Excess

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Abstract

This paper is the first in a separate working-paper line that uses Lagrangian Variable Cosmology (LVC) as a diagnostic lens on the dark-matter sector, distinct from the LVC working-paper series itself (v0.1–v0.3 phenomenology, v1.1 sine–Gordon reconstruction [1], v1.2 Coleman–Weinberg reconstruction [2]). The starting point is the empirical fact that the LVC modulation $H(z) = H_{\Lambda\text{CDM}}(z) \cdot T(z)$ fits the combined BAO + Planck distance-prior + SH0ES + Pantheon+ data set with $\Delta\text{BIC} \approx -30$ on $N = 1738$. Granted that fit as an empirical condition, this paper does not advocate the LVC ansatz; it asks what follows for the matter sector *if* the ansatz holds. Four results are recorded. First, under the field redefinition $T(z) \mapsto \Omega_m(z)$ obtained by writing $H^2(z) = H_0^2[\Omega_m(z)(1+z)^3 + (1-\Omega_m(z))]$, the LVC multiplicative modulation is mathematically equivalent to an evolving matter-density curve $\Omega_m(z)$ that asymptotes to a single baseline $\Omega_{m,\text{base}}$ at $z = 0$ and at high redshift and undergoes a localised excursion in the modulation window. Second, in this coordinate the relation $\delta\rho_m(z) \cdot (1+z)^{-3} + \delta\rho_\Lambda(z) = 0$ holds identically for all z and for any modulation profile $T(z)$, with $\delta n \equiv \delta\rho_m \cdot (1+z)^{-3}$ the comoving number-density excess relative to the baseline. The identity is algebraic, coordinate-invariant, and independent of the specific Lagrangian realisation. Third, applied to the v1.1 half-twist sine–Gordon kink, the comoving excess takes the form $\delta n \propto j_{0,\text{top}}^2$ where $j_{0,\text{top}} = (4\pi)^{-1}\partial_\theta\vartheta_{\text{kink}}$ is the topological current density; the topological charge $Q = \int j_{0,\text{top}}\,\mathrm{d}\theta = 1$ is preserved while $\delta n(z)$ vanishes at both ends of the observed range and peaks in the kink interior. Applied to the v1.2 R6 derivative-of-Gaussian bounce, the corresponding relation is $\delta n \propto -\partial_u\bar{\phi}$, linear rather than quadratic in the field gradient; the field displacement $|\bar{\phi}(\infty) - \bar{\phi}(0)| = A/2$ is a continuous parameter, not a topologically quantised charge. Fourth, the two equivalent rewritings ("effective-source" vs. "baseline-source") of the linear growth equation make distinct percent-level predictions for $f\sigma_8(z)$ at redshift-space distortion measurement points, and the time-delay distance shifts at the seven TDCOSMO lens systems exhibit distinct sign patterns across modulation families. Five items are recorded as open in Section 6 as the explicit programme for subsequent papers in this line.

1 Introduction

The LVC working-paper series, in its phenomenological line (v1–v16) and its reconstruction line (v0.1–v0.3, v1.1, v1.2), has documented a multiplicative modulation $T(z) = H(z)/H_{\Lambda\text{CDM}}(z)$ of the late-time expansion rate that returns to unity at $z = 0$ and at high redshift, exhibits a localised excursion in the window $z \approx 0.6$–$\sqrt{2}$ depending on the parametrisation, and reduces the combined fit to BAO + Planck distance-prior + SH0ES (PP-excluded, $N = 36$) and BAO + Planck + SH0ES + Pantheon+ (PP-included, $N = 1738$) by $\Delta\text{BIC} = -26.41$ and -30.31 respectively against ΛCDM with $k = 3$ [1]. Two Lagrangian reconstructions of this profile have been recorded: a half-twist sine–Gordon kink whose squared quarter-angle is $\mathrm{sech}^2((\theta-\theta_c)/w)$ at $\theta_c = \ln\varphi$ [1], and an R6 derivative-of-Gaussian profile $T(z) = 1 + A(z/w)\exp[-(z/w)^2]$ recovered as the static BPS-saturated bounce of a Coleman–Weinberg type logarithmic potential at $w = 2$ [2].

The present paper does not extend the LVC programme. It does not propose a new modulation, a new Lagrangian, or a tighter lock. It asks a different question. Granting the empirical fact that the LVC ansatz $H = H_{\Lambda\mathrm{CDM}} \cdot T$ fits the combined data set at the levels quoted, what follows for the matter sector if the ansatz is taken at face value?

The question is phrased as a conditional. No claim is attached to the truth of the LVC ansatz beyond its documented empirical performance. The results below are statements of the form "if the LVC ansatz holds, then …". They are not contingent on the truth of any specific reconstruction (v1.1 or v1.2 or otherwise) except where explicitly indicated; the central identity (Section 3) is independent of the choice of Lagrangian.

The paper is organised as follows. Section 2 records the mathematical equivalence between the multiplicative modulation $T(z)$ and an evolving matter-density curve $\Omega_m(z)$. Section 3 records the conservation identity $\delta\rho_m \cdot (1+z)^{-3} + \delta\rho_\Lambda = 0$ as an algebraic consequence of that rewriting. Section 4 applies the identity to the v1.1 sine–Gordon kink and to the v1.2 R6 bounce, recording the distinct relation between the comoving particle excess δn and the underlying field-theoretic structure in the two cases. Section 5 records two percent-level observational signatures (linear-growth $f\sigma_8$ and time-delay distance $D_{\Delta t}$) that distinguish the two interpretations of the rewriting and the two soliton-class realisations. Section 6 records five items as the explicit programme for subsequent papers in this line.

The discipline of the paper is the same as in v1.1 and v1.2: phenomenology and reconstruction are kept separate from physical claims. The reconstruction is an inverse program. No claim is made beyond the consistency of the recorded identities with the previously fitted modulation.

2 The $\Omega_m(z)$ rewriting

The LVC ansatz is

$$H^2(z) = H_0^2 \cdot T^2(z) \cdot \left[\Omega_{m,\mathrm{base}}(1+z)^3 + (1-\Omega_{m,\mathrm{base}})\right], \tag{1}$$

with $T(z) \to 1$ at $z = 0$ and at high redshift, $\Omega_{m,\mathrm{base}}$ a single constant whose value is taken from the locked construction of v1.1 ($\Omega_{m,\mathrm{base}} = (1+\exp(3\ln\varphi/\varphi))^{-1} = 0.290653$ [1]) or from the free-fit value of v1.2 R6 (typically $\Omega_{m,\mathrm{base}} \approx 0.28$–$0.30$ [2]).

Define $\Omega_m(z)$ by requiring that the LVC Hubble rate take the ΛCDM functional form with a redshift-dependent matter fraction:

$$\frac{H^2(z)}{H_0^2} = \Omega_m(z)\,(1+z)^3 + (1-\Omega_m(z)). \tag{2}$$

Equating (1) with (2) and solving for $\Omega_m(z)$ gives

$$\Omega_m(z) = \frac{T^2(z)\left[\Omega_{m,\mathrm{base}}(1+z)^3 + (1-\Omega_{m,\mathrm{base}})\right] - 1}{(1+z)^3 - 1}. \tag{3}$$

The right-hand side is well-defined for all $z > 0$. At $z = 0$ both numerator and denominator vanish; expanding to first order in z gives

$$\lim_{z\to 0^+} \Omega_m(z) = \Omega_{m,\mathrm{base}} + \frac{2T'(0)}{3}, \tag{4}$$

where $T'(0) \equiv \mathrm{d}T/\mathrm{d}z\big|_{z=0}$. For modulations satisfying $T(0) = 1$ and $T(\infty) = 1$, equation (3) returns $\Omega_m(z) \to \Omega_{m,\mathrm{base}}$ as $z \to \infty$ and to the value given by (4) as $z \to 0^+$.

Three numerical observations are recorded for the v1.1 sech^2 profile and the v1.2 R6 derivative-of-Gaussian profile, both at their lock values:

- Peak position. The $\Omega_m(z)$ curve attains its maximum at $z = 0.598$ for v1.1 (modulation peak $z = 1/\varphi = 0.618$) and at $z = 0.866$ for R6 (modulation peak $z = \sqrt{2} = 1.414$). The peak in $\Omega_m(z)$ is shifted from the peak in $T(z)$ by the kinematic factor $(1+z)^3 - 1$ in the denominator of (3).
- Peak amplitude. $\Omega_m(z)_{\mathrm{peak}} = 0.358$ for v1.1 (+23% above baseline) and 0.317 for R6 (+9% above baseline).
- Equality-epoch shift. In ΛCDM at $\Omega_m = 0.290653$, the matter–Λ equality occurs at $z_{\mathrm{eq}} = 0.346$. Under the $\Omega_m(z)$ rewriting, the equality $\rho_m(z) = \rho_\Lambda(z)$ holds at $z^*_{\mathrm{eq}} = 0.322$ (v1.1) and at $z^*_{\mathrm{eq}} = 0.293$ (R6). The shift is towards lower z in both cases.

Equation (3) is a coordinate rewriting, not a physical assumption. The two formulations (1) and (2) contain the same Friedmann equation; the redefinition shifts the location of the modulation between the prefactor $T(z)$ and the matter fraction $\Omega_m(z)$. The remainder of this paper records consequences of the rewriting that are not visible in the multiplicative form.

3 The conservation identity

Define, in the $\Omega_m(z)$ coordinate, the matter and dark-energy densities in units of the critical density $\rho_{c,0} \equiv 3H_0^2/(8\pi G)$:

$$\rho_m(z)/\rho_{c,0} = \Omega_m(z)\,(1+z)^3, \qquad \rho_\Lambda(z)/\rho_{c,0} = 1 - \Omega_m(z). \tag{5}$$

The baseline (ΛCDM) densities are

$$\rho_{m,\mathrm{base}}(z)/\rho_{c,0} = \Omega_{m,\mathrm{base}}(1+z)^3, \qquad \rho_{\Lambda,\mathrm{base}}/\rho_{c,0} = 1 - \Omega_{m,\mathrm{base}}. \tag{6}$$

The deviations from baseline at fixed z are

$$\delta\rho_m(z) \equiv \rho_m(z) - \rho_{m,\mathrm{base}}(z) = [\Omega_m(z) - \Omega_{m,\mathrm{base}}]\,(1+z)^3, \tag{7}$$

$$\delta\rho_\Lambda(z) \equiv \rho_\Lambda(z) - \rho_{\Lambda,\mathrm{base}} = -[\Omega_m(z) - \Omega_{m,\mathrm{base}}]. \tag{8}$$

Comparing (7) and (8) gives, identically in z and independently of the choice of $T(z)$,

$$\boxed{\delta\rho_m(z)\,(1+z)^{-3} + \delta\rho_\Lambda(z) = 0} \tag{9}$$

or equivalently $-\delta\rho_\Lambda/\delta\rho_m = (1+z)^{-3}$ wherever $\delta\rho_m \neq 0$. The identity (9) is algebraic. It is a consequence of the rewriting (2) together with the definitions (5)–(6); it does not invoke any equation of motion, conservation law, or property of $T(z)$ beyond its existence.

The quantity $\delta n(z) \equiv \delta\rho_m(z)\,(1+z)^{-3} = \Omega_m(z) - \Omega_{m,\mathrm{base}}$ has units of comoving number density (in units of $\rho_{c,0}$ per unit comoving volume). Equation (9) then reads $\delta n(z) = -\delta\rho_\Lambda(z)$. In words: under the rewriting (2), the comoving matter excess and the dark-energy deficit are equal in magnitude at every z, while the corresponding physical-density quantities $\delta\rho_m(z)$ and $\delta\rho_\Lambda(z)$ stand in the ratio $(1+z)^3 : -1$.

The identity has been verified numerically to machine precision for the v1.1 sech^2 profile and the v1.2 R6 derivative-of-Gaussian profile at the sample points

$$z \in \{0.05,\ 0.1,\ 0.2,\ 0.3,\ 0.5,\ 0.618,\ 0.8,\ 1.0,\ 1.414,\ 2.0,\ 3.0,\ 5.0,\ 10\}.$$

The absolute residuals on each side of (9) are below 10^{-15} at all sample points and for both modulations.

Coordinate invariance and a differential form are recorded as follows. Under the change of variable $\theta = \ln(1+z)$, the function $\Omega_m(\theta) = \Omega_m(z(\theta))$ takes the same numerical values; (9) therefore holds in θ-coordinate with the same form. Differentiating both sides of (9) with respect to z gives $\mathrm{d}\delta n/\mathrm{d}z + \mathrm{d}\delta\rho_\Lambda/\mathrm{d}z = 0$, which has been verified numerically to the same precision. Inclusion of the Hubble-friction correction discussed in Section 4.4 of [1] shifts the profile $T(z)$ at the percent level inside the modulation window but does not affect (9): the identity is algebraic and holds for any $T(z)$, regardless of whether $T(z)$ solves the field equation with or without friction.

The closed-form inverse of (9), expressing $T(z)$ in terms of $\delta n(z)$ and the baseline cosmology, follows by inverting (3):

$$T(z) = \sqrt{1 + \frac{\delta n(z)\,[(1+z)^3 - 1]}{\Omega_{m,\mathrm{base}}(1+z)^3 + (1-\Omega_{m,\mathrm{base}})}}. \tag{10}$$

Equation (10) is unique up to the choice of branch $T(z) > 0$; the modulation function and the comoving matter excess thus determine each other algebraically once the baseline cosmology is fixed.

4 Soliton-charge structure of the particle excess

The conservation identity (9) is universal. The relation between $\delta n(z)$ and the underlying field-theoretic data is not. This section records the explicit decomposition of $\delta n(z)$ in terms of the field-theoretic content of the two Lagrangian reconstructions in the LVC working-paper series.

4.1 v1.1: quadratic-in-current relation

The v1.1 reconstruction [1] writes the modulation as the static one-soliton sector of a half-twist sine–Gordon scalar ϑ with $V(\vartheta) = m^2[1 - \cos(\vartheta/2)]$. The kink solution is

$$\vartheta_{\mathrm{kink}}(\theta) = 8\arctan\bigl[\exp((m/2)(\theta - \theta_c))\bigr], \tag{11}$$

with $\theta = \ln(1+z)$ and the identifications $A = \varepsilon^2$, $w = 2/m$ giving the modulation

$$v(\theta) \equiv T(\theta) - 1 = A\,\sin^2(\vartheta_{\mathrm{kink}}/4). \tag{12}$$

The topological current density in the θ-coordinate is

$$j_{0,\mathrm{top}}(\theta) = \frac{1}{4\pi}\frac{\mathrm{d}\vartheta_{\mathrm{kink}}}{\mathrm{d}\theta} = \frac{1}{\pi w}\,\mathrm{sech}\bigl((\theta-\theta_c)/w\bigr), \tag{13}$$

with conserved charge

$$Q = \int_{-\infty}^{\infty} j_{0,\mathrm{top}}(\theta)\,\mathrm{d}\theta = \frac{\vartheta_{\mathrm{kink}}(\infty) - \vartheta_{\mathrm{kink}}(-\infty)}{4\pi} = 1. \tag{14}$$

From (12), the kink-sector identity $\sin(\vartheta_{\mathrm{kink}}/4) = \mathrm{sech}((\theta-\theta_c)/w)$, and (13), one obtains

$$v(\theta) = A\,(\pi w)^2\, j_{0,\mathrm{top}}^2(\theta). \tag{15}$$

The identity (15) has been verified numerically to machine precision at the sample points

$$\theta \in \{0.1,\, 0.2,\, 0.3,\, 0.481,\, 0.5,\, 0.6,\, 0.7\},$$

with $A = (\ln\varphi)^4$ and $w = (\ln\varphi)^3$ at the v1.1 lock values. Substituting (15) into the leading-order expansion of the conservation identity gives, in the small-amplitude limit $v \ll 1$,

$$\delta n(z) \simeq 2A(\pi w)^2\, j_{0,\mathrm{top}}^2(\theta)\, \frac{E^2_{\Lambda\mathrm{CDM}}(z)}{(1+z)^3 - 1}, \tag{16}$$

with $E^2_{\Lambda\mathrm{CDM}}(z) = \Omega_{m,\mathrm{base}}(1+z)^3 + (1 - \Omega_{m,\mathrm{base}})$. The leading correction to (16) is $O(A^2)$ and is given exactly by (3).

Equation (16) records the v1.1 structure. The topological charge $Q = 1$ is preserved exactly along the kink. The local density $j_{0,\mathrm{top}}(\theta)$ peaks at the kink centre $\theta = \theta_c$ and falls off as $\mathrm{sech}((\theta - \theta_c)/w)$ at both ends. The comoving particle excess $\delta n(z)$ is proportional to the squared density of this topological current. Both ends of the observed range ($z \to 0$ and $z \to \infty$) have $j_{0,\mathrm{top}} \to 0$ and hence $\delta n \to 0$ to leading order in A.

4.2 R6: linear-in-gradient relation

The R6 reconstruction [2] writes the modulation as the static BPS-saturated bounce of a bounded scalar $\bar\phi$ with the Coleman–Weinberg type logarithmic potential $V(\bar\phi) = 2\bar\phi^2 \ln(2\bar\phi/A)$. The static solution is

$$\bar\phi_{\mathrm{static}}(u) = \frac{A}{2}\, \exp(-u^2), \qquad u \equiv z/w, \tag{17}$$

and the modulation deviation is recovered as the negative gradient

$$v(u) \equiv T(u) - 1 = -\frac{\mathrm{d}\bar\phi}{\mathrm{d}u} = A\, u\, \exp(-u^2). \tag{18}$$

The field range is $\bar\phi \in [0, A/2]$, an interval rather than a compact phase. The field displacement

$$\Delta\bar\phi = |\bar\phi(\infty) - \bar\phi(0)| = A/2 \tag{19}$$

is a continuous parameter set by the amplitude A, not a topologically quantised charge.

The analogue of (15) for the R6 reconstruction is read off directly from (18):

$$v(u) = -\frac{\mathrm{d}\bar\phi}{\mathrm{d}u} = -w\, \frac{\mathrm{d}\bar\phi}{\mathrm{d}z}. \tag{20}$$

The corresponding comoving particle excess, to leading order in A, is

$$\delta n(z) \simeq -2w\, \frac{\mathrm{d}\bar\phi}{\mathrm{d}z}\, \frac{E^2_{\Lambda\mathrm{CDM}}(z)}{(1+z)^3 - 1}. \tag{21}$$

Equations (16) and (21) record the structural difference between the two reconstructions. The v1.1 relation is *quadratic* in the topological current density; the R6 relation is *linear* in the scalar-field gradient. The two relations do not interpolate: a sech^2 profile cannot be written as $-\partial_u\bar\phi$ for any $\bar\phi$ admitting the logarithmic potential, and a derivative-of- Gaussian profile cannot be written as $\sin^2(\vartheta/4)$ for any sine–Gordon kink. This structural difference is the precise form of the "classical/topological vs. quantum/effective" distinction stated in Section 8 of [2].

4.3 Asymmetry observation

Equations (16) and (21) share the same kinematic factor $E^2_{\Lambda\mathrm{CDM}}(z)/[(1+z)^3 - 1]$ from (9), and differ in the field-theoretic moment that multiplies it: $j^2_{0,\mathrm{top}}$ in v1.1 and $-\partial_u\bar\phi$ in R6. The origin of the asymmetry between the squared form in v1.1 and the linear form in R6 is not derived in this paper. It is recorded as a structural fact about the two specific reconstructions documented in [1, 2]. Whether the quadratic relation (16) is a feature of topologically quantised solitons in general (as opposed to the continuous bounce of R6) is not established here and is listed in Section 6.

z	Effective source			Baseline source		
	ΛCDM	v1.1 (Δ%)	R6 (Δ%)	ΛCDM	v1.1 (Δ%)	R6 (Δ%)
0.15	0.4426	+2.94	+3.42	0.4426	−0.31	−0.93
0.32	0.4624	+2.72	+2.29	0.4624	−0.80	−1.48
0.38	0.4655	+2.10	+1.92	0.4655	−1.34	−1.65
0.51	0.4665	−1.08	+1.18	0.4665	−3.66	−1.95
0.57	0.4649	−2.82	+0.86	0.4649	−4.64	−2.06
0.61	0.4632	−3.56	+0.66	0.4632	−4.83	−2.12
0.70	0.4579	−3.55	+0.23	0.4579	−3.69	−2.25
0.85	0.4459	−1.60	−0.40	0.4459	−0.74	−2.37
1.00	0.4312	−0.64	−0.93	0.4312	+0.51	−2.39
1.48	0.3794	−0.25	−1.96	0.3794	+1.00	−1.90

Table 1: Linear-growth $f\sigma_8(z)$ predictions under the LVC v1.1 and R6 locks, against ΛCDM with $\Omega_m = 0.290653$ and $\sigma_8(0) = 0.811$. "Effective source" uses the matter density $\Omega_m(z)(1+z)^3\rho_{c,0}$ from the rewriting (5); "baseline source" uses the fixed matter density $\Omega_{m,\mathrm{base}}(1+z)^3\rho_{c,0}$. The percent shifts are relative to ΛCDM at the same z.

5 Observational signatures

Two channels are recorded in which the rewriting of Section 2 and the soliton-charge structure of Section 4 produce percent-level deviations from ΛCDM that are in principle resolvable by current and near-future data.

5.1 Linear-growth signature: $f\sigma_8(z)$

The linear-growth equation for the matter density contrast δ in the subhorizon limit reads, in $N \equiv \ln a$,

$$\frac{\mathrm{d}^2\delta}{\mathrm{d}N^2} + \left(2 + \frac{\mathrm{d}\ln H}{\mathrm{d}N}\right)\frac{\mathrm{d}\delta}{\mathrm{d}N} = \frac{3}{2}\,\frac{\rho_{m,\mathrm{src}}(z)}{H^2(z)\,\rho_{c,0}^{-1}}\,\delta. \tag{22}$$

Under the LVC ansatz, $H(z)$ is given by (1) and the friction coefficient $\mathrm{d}\ln H/\mathrm{d}N$ is determined unambiguously by $T(z)$ and $\Omega_{m,\mathrm{base}}$. The source term, however, contains the matter density $\rho_{m,\mathrm{src}}(z)$, whose identification depends on whether the modulation is interpreted as a pure $H(z)$ modification (baseline source: $\rho_{m,\mathrm{src}} = \Omega_{m,\mathrm{base}}(1+z)^3\rho_{c,0}$) or as an evolving matter density (effective source: $\rho_{m,\mathrm{src}} = \Omega_m(z)(1+z)^3\rho_{c,0}$ from (5)). The two choices coincide at the background level (both give the same $H(z)$) but produce distinct growth predictions.

Table 1 records $f\sigma_8(z)$ at standard RSD measurement redshifts under the two source choices, for both the v1.1 sech^2 lock and the v1.2 R6 lock, against ΛCDM at $\Omega_m = \Omega_{m,\mathrm{base}} = 0.290653$ and a normalisation $\sigma_8(0) = 0.811$. The growth equation was integrated by LSODA in $N = \ln a$ from $z = 50$ to $z = 0$ with matter-domination initial conditions $\delta \propto a$, $\mathrm{d}\delta/\mathrm{d}N = \delta$.

Three observations follow from Table 1.

Observation 1. Under the effective-source interpretation, v1.1 changes sign across the modulation window: it predicts $+2.94\%$ at $z = 0.15$ and -3.56% at $z = 0.61$. Under the baseline-source interpretation, v1.1 is negative across the modulation window with peak deviation -4.83% at the modulation centre.

Observation 2. R6 under the effective-source interpretation is positive at low z ($+3.42\%$ at $z = 0.15$) and crosses zero around $z = 0.85$ to become negative at higher z. Under the baseline-source interpretation, R6 is negative across the entire range.

Observation 3. The differences between the two source interpretations at the same z are at the level of 1–3%, comparable to current state-of-the-art RSD uncertainty. The two interpretations

Lens	Modulation	$\Delta D_C(0 \to z_l)/D_C$ %	$\Delta D_C(z_l \to z_s)/D_C$ %	$\Delta D_{\Delta t}$ %
PG1115+080	v1.1	−0.101	−1.576	+0.288
($z_l = 0.311, z_s = 1.722$)	R6	−0.569	−2.503	−0.056
HE0435−1223	v1.1	−0.427	−1.662	+0.036
($z_l = 0.456, z_s = 1.693$)	R6	−0.808	−2.685	−0.099

Table 2: Decomposition of the $D_{\Delta t}$ shift into contributions from the integration domains $[0, z_l]$ and $[z_l, z_s]$ for the two sign-opposite TDCOSMO lenses. Base parameters $H_0 = 70$, $\Omega_m = 0.30$.

of the LVC ansatz are therefore in principle distinguishable by precision $f\sigma_8(z)$ measurements; whether they are resolved by current data is not assessed in this paper.

5.2 Time-delay signature: $D_{\Delta t}$ sign patterns

The strong-lensing time-delay distance in a flat universe with multiplicative LVC modulation is

$$D_{\Delta t}(z_l, z_s) = (1 + z_l)\,\frac{D_A(z_l)\,D_A(z_s)}{D_A(z_l, z_s)}, \qquad D_A(z) = \frac{D_C(z)}{1+z}, \tag{23}$$

with $D_C(z) = \int_0^z c\,\mathrm{d}z'/H(z')$. Table 2 of [2] records the pure-modulation shifts in $D_{\Delta t}$ across the seven TDCOSMO lens systems at fixed base parameters $H_0 = 70$, $\Omega_m = 0.30$. The shifts have opposite signs at PG1115+080 and HE0435−1223 for the v1.1 and R6 reconstructions.

The origin of the sign opposition is recorded by decomposition of the $D_{\Delta t}$ shift into contributions from the integration domain $[0, z_l]$ and $[z_l, z_s]$:

$$\Delta D_{\Delta t}/D_{\Delta t} \simeq \frac{\Delta D_C(z_l)}{D_C(z_l)} + \frac{\Delta D_C(z_s)}{D_C(z_s)} - \frac{\Delta[D_C(z_s) - D_C(z_l)]}{D_C(z_s) - D_C(z_l)}, \tag{24}$$

where the last term arises from the angular-diameter distance between lens and source. The numerical decomposition is recorded in Table 2 for the two sign-opposite lenses.

The ratio of pre-lens to post-lens contributions to $\Delta D_C/D_C$ at PG1115+080 is $0.101/1.576 = 0.064$ for v1.1 (post-lens dominates strongly) and $0.569/2.503 = 0.227$ for R6 (post-lens still dominates but pre-lens is non-negligible). The denominator in (24) contains the difference $D_C(z_s) - D_C(z_l)$, which is affected primarily by the post-lens contribution; the numerator contains the product $D_C(z_l)D_C(z_s)$, which is affected by both. The sign of $\Delta D_{\Delta t}$ is therefore determined by which contribution dominates the decrease. The narrow v1.1 modulation concentrates its effect in $[z_l, z_s]$ for lenses with $z_l < 1/\varphi < z_s$, driving the denominator down faster than the numerator and yielding $\Delta D_{\Delta t} > 0$. The wider R6 modulation contributes appreciably to $[0, z_l]$ as well, partially cancelling the denominator effect and yielding $\Delta D_{\Delta t} < 0$.

The sharpness ratio A/FWHM_z of the modulation, with FWHM measured in z, takes the values 0.168 for v1.1, 0.014 for R6. The sign pattern across the seven TDCOSMO lenses is a function of this sharpness, the modulation peak location, and the (z_l, z_s) distribution of the lens sample. The current TDCOSMO seven-lens combined $D_{\Delta t}$ precision is approximately 3% [3]; the sign-opposite predictions at PG1115+080 (±0.06–0.29%) and HE0435−1223 (±0.04–0.10%) are below this precision and are not resolved by present data.

6 Limitations and the programme

The results recorded above are conditional on the LVC ansatz $H = H_{\Lambda\mathrm{CDM}} \cdot T$ and use the locked values from [1, 2]. Five items are recorded as the explicit programme for subsequent papers in this line.

Item A: Physical content of the rewriting. The rewriting (2) and the identity (9) are coordinate-level statements. Three physical interpretations are compatible with the algebra and are not distinguished by the present analysis: (a) actual dark-matter particle number is created and destroyed with z; (b) particle number is conserved while the effective particle mass varies with z; (c) the quantity called "dark-matter density" in the data analysis is an effective density distinct from a conserved particle number. The v1.1 result (16) excludes (a) within the v1.1 reconstruction: the topological charge Q is exactly conserved and the apparent excess $\delta n(z)$ is the squared density of a current rather than a number- density variation. Whether (b) or (c) or both are realised in v1.1 is not determined here. The corresponding question in R6 is left open.

Item B: Origin of the quadratic-versus-linear asymmetry. Equations (16) and (21) record a structural difference between v1.1 and R6: the comoving particle excess is quadratic in the topological current density in the kink case, linear in the field gradient in the bounce case. Whether this asymmetry is a general feature of topologically quantised soliton sectors (as opposed to continuous bounces), or specific to the two reconstructions documented in the LVC series, is not assessed here. A third reconstruction satisfying the four LVC requirements (L, C, A, P) of [2] Section 2 has been verified to be possible at the level of an existence statement (an S^2 sigma-model profile satisfying the universal identity (9) can be written down), but no Lagrangian-side reconstruction at the closure level of v1.1 or v1.2 has been produced for that case in the present paper.

Item C: Early–late matter separation. The Planck distance prior involves the matter density at recombination ($z \sim 1090$), where $T(z) \to 1$ and $\Omega_m(z) \to \Omega_{m,\mathrm{base}}$, while the BAO and SN data involve the matter density in the modulation window ($z \lesssim 5$), where $\Omega_m(z)$ departs from $\Omega_{m,\mathrm{base}}$. The two channels nominally measure "the same Ω_m" under the standard reading. Under the rewriting of Section 2, the two values are formally distinct functionals of $T(z)$. Whether the data admit, prefer, or reject a separation between early and late Ω_m under the LVC ansatz has not been tested by an explicit fit in this paper and is the natural quantitative continuation.

Item D: Effective vs. baseline source. The two interpretations of the source term in (22) were treated as alternatives in Table 1. Within the LVC reconstructions of [1, 2], no internal principle of the action selects between them; the effective-source interpretation is consistent with the $\Omega_m(z)$ rewriting at the level of the background but requires an additional specification at the perturbation level. A first-principles determination of the source term from the v1.1 or R6 Lagrangian, including the coupling of ϑ or $\bar{\phi}$ to matter perturbations, is not given here.

Item E: Comparison with real $f\sigma_8(z)$ and $D_{\Delta t}$ measurements. Tables 1 and 2 record predictions. A confrontation with the compiled $f\sigma_8(z)$ measurements from BOSS, eBOSS, and DESI, and with the lens-specific $D_{\Delta t}$ posteriors of TDCOSMO, has not been performed. The question of whether current data prefer the effective-source or the baseline-source interpretation, or distinguish v1.1 from R6 (or from a third reconstruction satisfying the four LVC requirements), is the empirical continuation of the present paper.

7 Outlook

The present paper records four results conditional on the LVC ansatz. The $\Omega_m(z)$ rewriting (Section 2) makes the LVC multiplicative modulation algebraically equivalent to a redshift-dependent matter density that returns to a fixed baseline at $z = 0$ and at high redshift. The conservation identity (9) (Section 3) relates the comoving matter excess and the dark-energy deficit at every z and for any choice of $T(z)$. The soliton-charge decomposition (Section 4) records a quadratic-in-current relation in v1.1 and a linear-in-gradient relation in R6, distinguishing the two reconstructions documented in the LVC series at the level of how the field-theoretic content maps to the comoving particle excess. The observational signatures (Section 5) record percent-level predictions for $f\sigma_8(z)$ and sign patterns for $D_{\Delta t}$ that distinguish the two source interpretations and the two reconstructions.

Whether any of these structures reflects a physical feature of the dark- matter sector, or whether the agreement between the LVC ansatz and the data admits multiple interpretations of which only some involve a non-trivial matter-sector statement, is not adjudicated by the present paper. The five limitations recorded in Section 6 constitute the explicit programme for the subsequent papers in this line.

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References

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